EMPLOYEE ANALYTICS :

STRUCTURE :

MASTER TABLES :

DEPARTMENTS TABLE :

```
create table departments
(
dept_id int primary key,
dept_name varchar(50) not null
)
```

LOCATION TABLE :

```
create table locations(location_id int primary key,
location_name varchar(50) not null)
```

JOB ROLE TABLE :

```
create table job_role(
role_id int primary key,
role_name varchar(50) not null,
minsalary decimal(10,2),
maxsalary decimal(10,2)
)
```

EMPLOYEES TABLE :

```
create table employees(
emp_id int primary key,
emp_name varchar(100) not null,
gender varchar(10),
dept_id int,
role_id int,
location_id int,
join_date date,
status varchar(20) default 'Active',
foreign key (dept_id) references departments(dept_id),
foreign key (role_id) references job_role(role_id),
foreign key (location_id) references locations(location_id)
)
```

SALARY TABLE :

```
create table salaries(
salary_id int primary key,
emp_id int,
salary decimal(10,2),
effective_date date,
foreign key (emp_id) references employees(emp_id)
)
```

PERFORMANCE TABLE

```
create table perfomance_reviews(
review_id int primary key,
emp_id int ,
review_year int,
rating decimal(2,1) check (rating between 1 and 5),
foreign key (emp_id) references employees(emp_id)
)
```

INSERTION :

```
insert into departments values
(1,'IT'),
(2,'HR'),
(3,'Finance'),
(4,'Marketing'),
(5,'Operations')

insert into locations values
(1,'Bangalore’'),
(2,'Pune'),
(3,'Hyderabad')

INSERT INTO job_role (role_id, role_name, minsalary, maxsalary)
VALUES
(1, 'Software Engineer', 40000.00, 90000.00),
(2, 'Data Analyst', 35000.00, 75000.00),
(3, 'Project Manager', 50000.00, 120000.00),
(4, 'HR Specialist', 30000.00, 70000.00),
(5, 'Marketing Executive', 32000.00, 85000.00),
(6, 'Sales Manager', 45000.00, 110000.00),
(7, 'System Administrator', 38000.00, 80000.00),
(8, 'Business Analyst', 42000.00, 95000.00),
(9, 'UX Designer', 37000.00, 88000.00),
```

```
(10, 'Finance Officer', 40000.00, 100000.00);
```

BULK CSV FILE LOADING (DATASET) :

EMPLOYEES FILE LOADING :

```
BULK INSERT employees
FROM 'C:\Program Files\Microsoft SQL
Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\Employees_100_Records.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
    TABLOCK,
    CODEPAGE = '65001'

);
```

SALARIES FILE LOADING :

```
BULK INSERT salaries
FROM 'C:\Program Files\Microsoft SQL
Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\salaries.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
    TABLOCK,
    CODEPAGE = '65001'

);
```

PERFORMANCE_REVIEWS FILE LOADING :

```
BULK INSERT perfomance_reviews
FROM 'C:\Program Files\Microsoft SQL
Server\MSSQL16.MSSQLSERVER\MSSQL\DATA\perfomance_reviews.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
    TABLOCK,
    CODEPAGE = '65001'

);
```

PERFORMANCE ANALYTICS :

1.TOP 10 HIGH PERFORMING EMPLOYEES :

```
select top 10 * from employees e join salaries s on
e.emp_id=s.emp_id
join perfomance_reviews p on e.emp_id = p.emp_id order by p.rating
desc
```

Results | Messages

	emp_id	emp_name	gender	dept_id	role_id	location_id	join_date	status	salary_id	emp_id	salary	effective_date	review_id	emp_id	review_year	rating
1	6	Priya Sharma 6	Male	4	1	2	2024-07-03	Inactive	6	6	52500.00	2020-01-01	6	6	2020	4.7
2	16	Divya Patel 16	Female	5	4	2	2017-06-17	Active	16	16	57500.00	2020-01-01	16	16	2021	4.7
3	26	Sneha Iyer 26	Male	3	4	3	2022-04-15	Inactive	26	26	62500.00	2020-01-01	26	26	2022	4.7
4	36	Priya Sharma 36	Male	2	4	2	2020-05-25	Active	36	36	67500.00	2020-01-01	36	36	2023	4.7
5	46	Priya Sharma 46	Male	5	3	1	2018-11-30	Active	46	46	72500.00	2020-01-01	46	46	2024	4.7
6	56	Ananya Rao 56	Male	2	1	2	2023-10-01	Inactive	56	56	77500.00	2020-01-01	56	56	2025	4.7
7	66	Rohit Das 66	Male	3	4	1	2022-06-13	Inactive	66	66	82500.00	2020-01-01	66	66	2026	4.7
8	76	Priya Sharma 76	Female	2	2	3	2020-10-22	Inactive	76	76	87500.00	2020-01-01	76	76	2027	4.7
9	86	Rohit Das 86	Male	4	2	2	2019-11-21	Inactive	86	86	92500.00	2020-01-01	86	86	2028	4.7
10	96	Karthik R 96	Male	1	5	3	2024-04-25	Active	96	96	97500.00	2020-01-01	96	96	2029	4.7

2. AVERAGE RATING PER DEPARTMENT :

```
SELECT AVG(p.rating) as Average_Rating, d.dept_name
FROM perfomance_reviews p
JOIN employees e ON e.emp_id = p.emp_id
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Results | Messages

	Average_Rating	dept_name
1	3.491304	Finance
2	3.661904	HR
3	3.844444	IT
4	3.538095	Marketing
5	3.664705	Operations

Query executed successfully.

3. EMPLOYEES ELIGIBLE FOR PROMOTION (RATING > 4 & 2+ YEARS)

```
SELECT e.emp_id, e.emp_name
FROM employees e
JOIN perfomance_reviews p ON e.emp_id = p.emp_id
WHERE p.rating > 4
  AND e.emp_id IN (
      SELECT emp_id
      FROM employees
      WHERE Datediff(YEAR, join_date, GETDATE()) >= 2
  );
```

Results | Messages

	emp_id	emp_name
1	4	Rohit Das 4
2	6	Priya Shar...
3	9	Arun Kum...
4	14	Priya Shar...
5	16	Divya Pat...
6	19	Rohit Das...
7	24	Sneha Iye...
8	26	Sneha Iye...
9	29	Sneha Iye...
10	34	Rohit Das...

Query executed successfully.

4. PERFORMANCE TRENDS OVER YEARS :

```
  SELECT p.review_year,
       AVG(p.rating) AS avg_rating,
       COUNT(*) AS total_reviews
FROM perfomance_reviews p
GROUP BY p.review_year
ORDER BY p.review_year;
```

Results | Messages

	review_year	avg_rating	total_reviews
1	2020	3.630000	10
2	2021	3.630000	10
3	2022	3.630000	10
4	2023	3.630000	10
5	2024	3.630000	10
6	2025	3.630000	10
7	2026	3.630000	10
8	2027	3.630000	10
9	2028	3.630000	10
10	2029	3.630000	10

Query executed successfully.

-- Salary Analytics

5. SALARY DISTRIBUTIONS PER DEPARTMENTS :

```
select d.dept_name,sum(s.salary) as Salary_Distribution from
employees e
join departments d on e.dept_id = d.dept_id
join salaries s on e.emp_id = s.emp_id
Group by d.dept_name
```

Results | Messages

	dept_name	Salary_Distribution
1	Finance	1768000.00
2	HR	1617500.00
3	IT	1310500.00
4	Marketing	1607500.00
5	Operations	1171500.00

6. HIGHEST AND LOWEST SALARY PER ROLE

```
select jr.role_name,min(s.salary) as minimum_salary,max(s.salary) as
maximum_salary from salaries s
join employees as e on s.emp_id = e.emp_id
join job_role as jr on e.role_id = jr.role_id
group by jr.role_name
order by jr.role_name
```

Results | Messages

	role_name	minimum_salary	maximum_salary
1	Data Analyst	50000.00	96000.00
2	HR Specialist	50500.00	94500.00
3	Marketing E...	51000.00	99500.00
4	Project Man...	52000.00	96500.00
5	Software E...	52500.00	99000.00

7. SALARY VS PERFORMANCE CORRELATION :

```
select e.emp_id ,e.emp_name,avg(pr.rating) as avg_rating ,
s.salary from employees e
join perfomance_reviews as pr
on e.emp_id=pr.emp_id
join salaries s
on e.emp_id = s.emp_id
group by e.emp_id,e.emp_name,s.salary
order by avg_rating desc
```

Results | Messages

	emp_id	emp_name	avg_rating	salary
1	6	Priya Sharma 6	4.700000	52500.00
2	16	Divya Patel 16	4.700000	57500.00
3	26	Sneha Iyer 26	4.700000	62500.00
4	36	Priya Sharma 36	4.700000	67500.00
5	46	Priya Sharma 46	4.700000	72500.00
6	56	Ananya Rao 56	4.700000	77500.00
7	66	Rohit Das 66	4.700000	82500.00
8	76	Priya Sharma 76	4.700000	87500.00
9	86	Rohit Das 86	4.700000	92500.00
10	96	Karthik R 96	4.700000	97500.00
11	94	Divya Patel 94	4.500000	96500.00

Query executed successfully.

8. EMPLOYEES BELOW DEPARTMENT AVERAGE SALARY :

```
select
    e.emp_id,
    e.emp_name,
    d.dept_name,
    s.salary
from employees e
join salaries s on e.emp_id = s.emp_id
join departments d on e.dept_id = d.dept_id
```

```
where s.salary < (
    select avg(s2.salary)
    from employees e2
    join salaries s2 on e2.emp_id = s2.emp_id
    where e2.dept_id = e.dept_id
);
```

Results | Messages

	emp_id	emp_name	dept_name	salary
1	7	Divya Patel 7	IT	53000.00
2	11	Sneha Iyer ...	IT	55000.00
3	17	Rohit Das 17	IT	58000.00
4	19	Rohit Das 19	IT	59000.00
5	23	Meera Nair ...	IT	61000.00
6	24	Sneha Iyer ...	IT	61500.00
7	27	Meera Nair ...	IT	63000.00
8	29	Sneha Iyer ...	IT	64000.00
9	40	Ananya Ra...	IT	69500.00
10	41	Arun Kumar...	HR	70000.00
11	30	Vikram Sin...	HR	64500.00
12	34	Rohit Das 34	HR	66500.00

Query executed successfully.

DEPARTMENT GROWTH :

9. HIRING TREND PER YEAR :

```
SELECT YEAR(join_date) AS hire_year,
       COUNT(*) AS hires
FROM employees
GROUP BY YEAR(join_date)
ORDER BY hire_year;
```

Results | Messages

	hire_year	hires
1	2015	15
2	2016	9
3	2017	11
4	2018	10
5	2019	9
6	2020	10
7	2021	7
8	2022	11
9	2023	8
10	2024	10

Query executed successfully.

10. HEADCOUNT GROWTH BY DEPARTMENTS :

```
SELECT d.dept_name,
       YEAR(e.join_date) AS hire_year,
       COUNT(*) AS hires_in_year
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name, YEAR(e.join_date)
ORDER BY d.dept_name, hire_year;
```

Results | Messages

	dept_name	hire_year	hires_in_year
1	Finance	2015	2
2	Finance	2016	2
3	Finance	2017	4
4	Finance	2019	3
5	Finance	2020	1
6	Finance	2021	2
7	Finance	2022	5
8	Finance	2024	4
9	HR	2015	5
10	HR	2017	2
11	HR	2018	5
12	HR	2019	2

Query executed successfully.

11. ATTRITION RATE BY DEPARTMENT :

```
SELECT d.dept_name,
       COUNT(CASE WHEN e.status = 'Inactive' THEN 1 END) * 1.0 /
COUNT(*) AS attrition_rate
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Results | Messages

	dept_name	attrition_rate
1	Finance	0.434782608695
2	HR	0.666666666666
3	IT	0.500000000000
4	Marketing	0.523809523809
5	Operations	0.176470588235

STRATEGIC INSIGHTS

12. EMPLOYEES OVERDUE DUE PROMOTION :

```
SELECT e.emp_id, e.emp_name, e.join_date,
       DATEDIFF(YEAR, e.join_date, GETDATE()) AS experience_years
         FROM employees e
         WHERE DATEDIFF(YEAR, e.join_date, GETDATE()) >= 3;
```

Results | Messages

	emp_id	emp_name	join_date	experience_years
1	1	Priya Sharma 1	2016-07-25	10
2	2	Rohit Das 2	2015-05-03	11
3	3	Sneha Iyer 3	2021-04-17	5
4	4	Rohit Das 4	2018-02-13	8
5	5	Rahul Verma 5	2017-05-31	9
6	7	Divya Patel 7	2016-05-26	10
7	8	Priya Sharma 8	2017-02-26	9
8	10	Karthik R 10	2019-02-25	7
9	11	Sneha Iyer 11	2016-12-01	10
10	12	Rahul Verma...	2022-09-19	4
11	13	Ananya Rao ...	2019-07-02	7
12	14	Priya Sharma...	2022-05-08	4

Query executed successfully.

13. HIGH SALARY BUT LOW PERFORMANCE CASES :

```
SELECT e.emp_id, e.emp_name, s.salary, p.rating
FROM employees e
JOIN perfomance_reviews p ON e.emp_id = p.emp_id
JOIN salaries s ON e.emp_id = s.emp_id
WHERE s.salary > 80000.0
  AND p.rating < 3.0;
```

Results | Messages

	emp_id	emp_name	salary	rating
1	63	Arun Kumar 63	81000.00	2.8
2	68	Ananya Rao 68	83500.00	2.5
3	73	Vikram Singh 73	86000.00	2.8
4	78	Rohit Das 78	88500.00	2.5
5	83	Karthik R 83	91000.00	2.8
6	88	Rahul Verma 88	93500.00	2.5
7	93	Arun Kumar 93	96000.00	2.8
8	98	Ananya Rao 98	98500.00	2.5

14. MOST STABLE DEPARTMENTS(LOWEST ATTRITION) :

```
SELECT d.dept_name,
COUNT(CASE WHEN e.status = 'Inactive' THEN 1 END) * 1.0 / COUNT(*)
AS attrition_rate
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
order by attrition_rate asc

SELECT top 1 d.dept_name,
COUNT(CASE WHEN e.status = 'Inactive' THEN 1 END) * 1.0 / COUNT(*)
AS attrition_rate
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
order by attrition_rate asc
```

Results | Messages

	dept_name	attrition_rate
1	Operations	0.176470588235

15. LOCATION-WISE SALARY COMPARISONS :

```
SELECT l.location_name,
       AVG(s.salary) AS avg_salary,
       MIN(s.salary) AS min_salary,
       MAX(s.salary) AS max_salary,
       COUNT(*) AS employee_count
FROM employees e
Join salaries s on e.emp_id = s.emp_id
Join locations l on e.location_id = l.location_id
Group by l.location_name
Order by avg_salary desc
```

Results | Messages

	location_name	avg_salary	min_salary	max_salary	employee_count
1	Pune	78120.689655	52500.00	99000.00	29
2	Hyderabad	76902.777777	51500.00	99500.00	36
3	Bangalore´	69742.857142	50000.00	97000.00	35

16. GENDER DIVERSITY RATIO :

```
SELECT gender,
       COUNT(*) AS total_employees,
       COUNT(*) * 100.0 / (SELECT COUNT(*) FROM employees) AS
percentage
FROM employees
GROUP BY gender;
```

Results | Messages

	gender	total_employees	percentage
1	Female	48	48.000000000000
2	Male	52	52.000000000000

17. MEDIAN SALARY CALCULATION :

```
SELECT distinct PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY salary)
over() AS median_salary
FROM salaries;
```

Results | Messages

	median_salary
1	74750

18. IDENTITY LEADERSHIP PIPELINES CANDIDATES :

```
select
    e.emp_id,
    e.emp_name,
    jr.role_name,
    avg(pr.rating) as avg_rating,
    datediff(year, e.join_date, getdate()) as experience_years
    from employees e
    join performance_reviews pr on e.emp_id = pr.emp_id
    join job_roles jr on e.role_id = jr.role_id
    group by e.emp_id, e.emp_name, jr.role_name, e.join_date
```

```
having avg(pr.rating) >= 4.5
and datediff(year, e.join_date, getdate()) >= 5;
```

Results | Messages

	emp_id	emp_name	role_name	avg_rating	experience_years
1	44	Vikram Singh 44	Data Analyst	4.500000	7
2	76	Priya Sharma 76	Data Analyst	4.700000	6
3	86	Rohit Das 86	Data Analyst	4.700000	7
4	4	Rohit Das 4	HR Specialist	4.500000	8
5	16	Divya Patel 16	HR Specialist	4.700000	9
6	34	Rohit Das 34	HR Specialist	4.500000	8
7	36	Priya Sharma 36	HR Specialist	4.700000	6
8	74	Meera Nair 74	HR Specialist	4.500000	5
9	84	Sneha Iyer 84	Marketing ...	4.500000	11
10	46	Priya Sharma 46	Project Ma...	4.700000	8
11	64	Priya Sharma 64	Project Ma...	4.500000	5
12	94	Divya Patel 94	Project Ma...	4.500000	10
13	54	Priya Sharma 54	Software E...	4.500000	5

Query executed successfully.

DESCRIPTION :

To analyze employee, salary, and performance data to generate meaningful business insights. It evaluates promotion eligibility, attrition trends, hiring growth, and salary-performance patterns using advanced sql queries.overall, it demonstrates strong data analysis and data engineering skills through joins, aggregations, and analytical functions.